Solar Filament Instance Segmentation under Multi-Annotator Panoptic Evaluation

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***Abstract*—We study instance segmentation of solar filaments in 2048×2048 GONG H-alpha observations under the organiser's panoptic quality metric. We show that the 707 training images contain 1,154 annotator passes scored independently, that cross-annotator agreement bounds PQ at 0.320, and that the union-of-annotators oracle reaches 0.782 overall but only 0.601 on three-pass images. Our pipeline couples a YOLO11-seg detector trained on union targets with a U-Net semantic prior that re-thresholds each proposal; it improves local PQ from 0.366 to 0.399 and transfers to the public leaderboard (0.290→0.340). We report ten further attempts that raised detector metrics without raising PQ, and argue that under FP/FN-symmetric multi-annotator evaluation the remaining bottleneck is recall on single-pass images.**

I. Task and Metric

Given a 2048×2048 GONG H-alpha image, predict one mask per filament. The metric is panoptic quality with IoU>0.5 matching,

$$\mathrm{PQ} = \frac{\sum_{(g,p)\in TP} \mathrm{IoU}(g,p)}{|TP| + 0.5|FP| + 0.5|FN|}, \tag{1}$$

evaluated per annotator pass and pooled over the whole test set. Submissions are CSV files of (`filament_id`, `segmentation_rle`) and masks of one image may not overlap. Each of these three properties matters for the design that follows: the metric is instance-level, it is *not* one-to-one (every pair above threshold counts), and the platform enforces mutual exclusion.

II. What the Data Contains

The 707 training images carry **1,154 annotator passes**: 411 images with one pass, 145 with two, 151 with three; 8,199 annotations. Table I summarises the instance statistics. Because each pass is scored independently, a filament marked by one expert is a TP for that pass and an FP for every other pass of the same image, and FP and FN carry equal weight. Scoring one expert's labels against another expert's labels for the same physical image gives **PQ 0.320** (SQ 0.631, RQ 0.507) over 120 images; this is the annotation-noise ceiling, not a modelling artefact. Using the union of all expert instances of an image as a perfect prediction bounds any method. On our held-out split this oracle scores **0.782** (SQ 0.921, RQ 0.849), but the bound depends strongly on annotation count (Table II). Any split-to-split comparison is therefore only meaningful at fixed composition, and a public/private gap is expected when the public half has more multi-pass images.

TABLE I
MAGFiLO statistics used in this work.

quantity	value	notes
physical images (train)	707	1,154 passes
instances	8,199	7.1 per image
median mask area	1,088 px	0.026% of frame
instances $<$ 400 px	9.6%	hardest slice
median bbox long side	95 px	p99 = 509 px
chirality classes	3 used	Left/Right/Unident.

TABLE II
Union oracle by passes per image (validation split).

passes/image	images	oracle PQ	oracle RQ
1	66	1.000	1.000
2	16	0.733	0.809
3	24	0.601	0.718
all	106	0.782	0.849

III. Method

Figure 1 shows the pipeline. **Union targets.** Duplicate instances across passes of one physical image are merged (mask IoU>0.6, 6,941→5,810 instances), so the detector learns to cover every filament any expert considered real. Training on single-pass labels instead raises YOLO's mask mAP50 from 0.52 to 0.68 yet *lowers* PQ (0.370 vs 0.366): it buys precision with recall, and the metric weights both equally.

Detector. YOLO11s-seg at 1280 px, 100 epochs, SGD with cosine schedule, mosaic and scale augmentation, no colour jitter since the input is greyscale. The 601/106 split is grouped by physical image so no GONG observation appears on both sides.

Semantic prior and refinement. A U-Net (resnet34, ImageNet encoder, 1024 px, same union targets) predicts a full-resolution probability field. Each detector proposal is dilated by 6 px to define an ROI and re-thresholded against the U-Net field at $\theta = 0.85$; components below 200 px fall back to the original proposal. The step can only move a boundary toward image evidence, never invent or delete an instance, which is why it cannot inflate FP the way free-form component extraction does.

Mutual exclusion. Instances are assigned greedily by detector confidence. This constraint costs ≈0.018 PQ relative to unconstrained output and is applied in every number reported here.

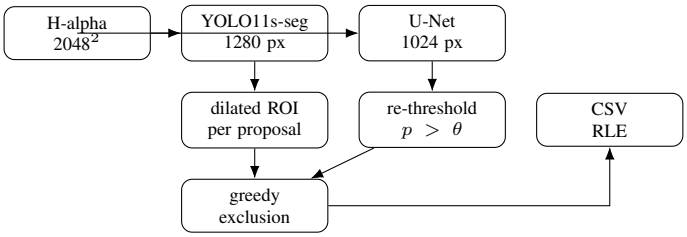


Fig. 1. YOLO proposals are refined against the U-Net semantic field inside a dilated ROI; instances are then made mutually exclusive.

IV. Related Work and Design Rationale

Solar filament segmentation has historically been approached with thresholding and morphological operators on contrast-enhanced H-alpha images, and more recently with U-Net-style semantic segmentation trained on the MAGFiLO annotations. Both families optimise pixel overlap (Dice/IoU), which is a different objective from the instance-level PQ used here: a semantic model that covers every filament region still has to *separate* touching filaments to score panoptic quality, and a detector that separates them still has to place each boundary within IoU 0.5 of an expert's trace.

Our design follows from that split. A detector supplies instance separation and a learned objectness prior; a semantic model supplies a full-resolution boundary field that the detector's low-resolution mask head cannot express (stride-4 prototypes upsampled to 2048 px). Refining rather than replacing was a deliberate choice: component extraction from the U-Net field alone reaches 0.382 and, when merged with proposals, 0.348 – the semantic model has good localisation but poor instance discrimination on touching filaments, whereas the detector has the opposite profile. Keeping the detector as the instance authority and the U-Net as the boundary authority combines their strengths; letting either dominate loses one of them.

The same reasoning explains the negative results. Flip TTA and tiled inference change the statistics the detector was trained on without adding information. Larger backbones and longer schedules improve average precision, but the misses we care about sit at IoU 0.27, just under the match threshold, and none of those changes moves them across it. Proposal re-ranking has the right idea – spend capacity on the decision boundary – but eight hand-designed features are too weak to separate near misses from false positives, and the held-out half shows the fit does not transfer.

V. Experiments

Table III lists every configuration we ran on the local split. Only U-Net refinement transfers to the leaderboard.

VI. Error Analysis and Limits

Splitting validation by annotation count locates the bottleneck (Table V). Mask quality is flat across groups (SQ≈0.67), but single-pass images lose 53% of instances to missed detections while three-pass images reach 73% of their (lower)

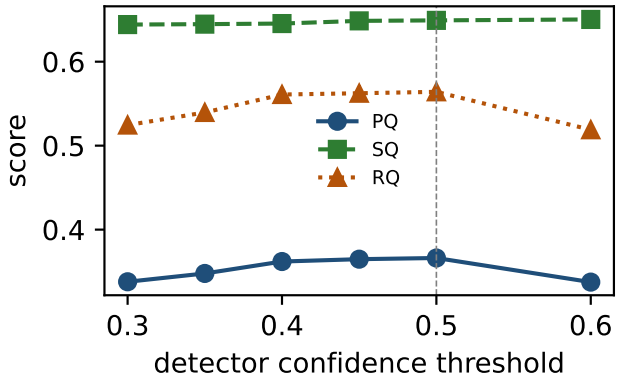


Fig. 2. Sensitivity to the detector confidence threshold (validation, refine pipeline). SQ keeps rising as low-confidence proposals are removed, but recall drops, and the product peaks at 0.50 – the value used for every reported result.

ceiling. We then asked whether misses are "not found" or "found badly". Matching every ground-truth instance to its best prediction regardless of the 0.5 threshold gives a mean best-IoU of **0.273** for missed instances: the detector overlaps them, but not enough to match. Recall by instance area is 60% below 400 px versus 74–78% above; recall by radial position is flat at 72–76% across four annuli, so limb darkening is not the cause. The flat radial profile and the near-threshold IoU together say the failure is boundary precision on faint objects, not a global contrast bias.

VII. Discussion

Two facts limit what model scaling can buy here. First, the metric is FP/FN-symmetric and evaluated against several annotators, so a prediction is rewarded for covering the union of expert opinion, not for agreeing with any single expert; this is why larger backbones and longer schedules that improve mAP50 do not move PQ. Second, 62% of validation images have a single pass whose oracle is 1.0, yet the pipeline reaches only 0.377 there: 53% of instances are missed outright. Improving recall on low-contrast, small filaments is the only lever that moves the aggregate, and it is bounded by 707 training images.

VIII. Submission Strategy

Three properties of the platform shaped the submission more than any model choice. First, masks must be mutually exclusive; we sort by confidence and give each pixel to its highest-scoring proposal, which costs 0.018 PQ but is mandatory. Second, the public leaderboard covers half the test set, so a local threshold chosen on 106 images can be a quarter-point off; we therefore selected the threshold by the shape of the PQ curve (Fig. 2), whose optimum is broad between 0.40 and 0.50, rather than by the single best point. Third, only one submission counts for final scoring, so there is no value in resubmitting near-identical variants to probe the public score; we used our submissions to validate two qualitatively different

TABLE III
ALL CONFIGURATIONS EVALUATED ON THE LOCAL VALIDATION SPLIT (OFFICIAL PQ, MUTUALLY EXCLUSIVE MASKS). "BASE" IS YOLO11S-SEG AT 1280 PX WITH UNION TARGETS UNLESS STATED.

configuration	key settings	PQ	note
base detector	YOLO11s, 100 ep, conf 0.50	0.366	reference
+ U-Net refine	$\theta = 0.85$, ROI 6 px, min-area 200	**0.399**	transferred to LB (0.340)
single-pass labels	one annotator per sample	0.370	mAP50 0.68 but recall loss
U-Net components merged	novel-iou 0.30, $\theta = 0.90$	0.348	added instances mostly FP
flip TTA	horizontal flip, mask-level dedup	0.371	filaments not mirror-symmetric
1536 inference	same weights, imgsz 1536	0.391	train/test scale mismatch
YOLO11m	60 ep, batch 8	0.396	higher mAP50, same PQ
s+m ensemble	proposal union then refine	0.373	FP 323 vs 156
tiled 1024	stride 768, crops fed at 1.25×	0.320	targets leave training scale
re-ranking	8 features, LR, held-out half	0.340	overfits fitting half
200-epoch training	early-stops at ep 133	0.393	mAP50 0.559, no PQ gain
contrast channels	$[I, \sigma_{64}, \sigma_{12}]$ input	0.395	recall up, PQ flat

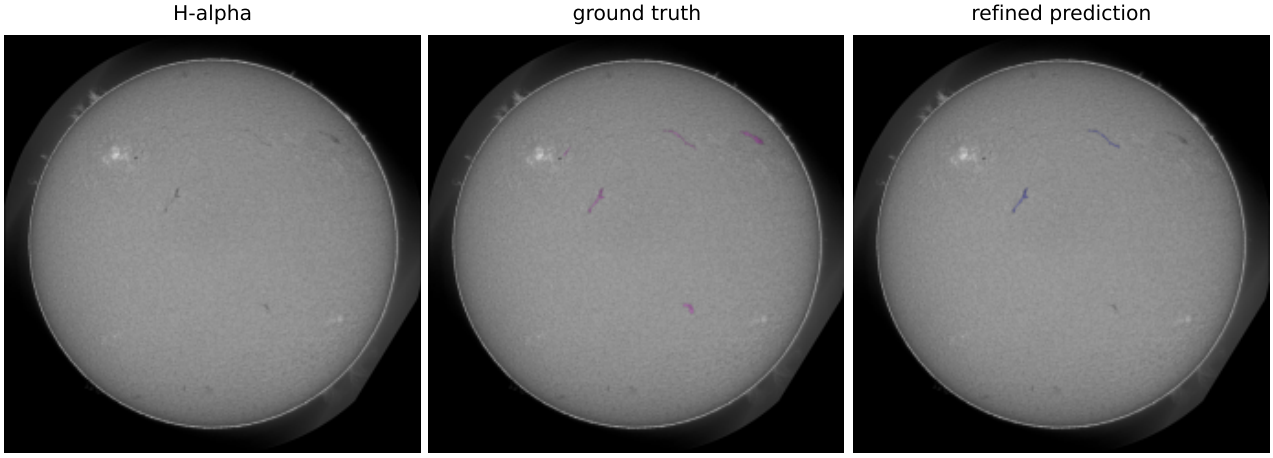


Fig. 3. Qualitative example. Left: H-alpha observation. Middle: one expert's ground-truth instance mask. Right: the refined prediction. The refined masks follow the faint elongated bodies and keep them separate; errors concentrate on the smallest, lowest-contrast fragments.

TABLE IV
REFINEMENT ABLATION (50 VALIDATION IMAGES). U-NET THRESHOLD DOMINATES; ROI DILATION IS NEARLY INERT ONCE IT COVERS THE BOUNDARY.

θ	ROI (px)	PQ	SQ
0.70	4	0.3868	0.6663
0.70	8	0.3843	0.6652
0.80	4	0.3943	0.6716
0.80	8	0.3927	0.6703
0.90	4	**0.3977**	0.6662
0.90	8	0.3972	0.6649
0.95	4	0.3850	0.6646

pipelines instead (base detector 0.290, refined 0.340), which confirmed that the refinement gain is real rather than a local-split artefact.

For inference we run the detector at the training resolution (1280 px) on the full 2048 px frame, take the masks at full resolution, refine each against the U-Net field computed from a 1024 px pass of the same image, and write compressed COCO RLE. The whole test split takes under four minutes on a single RTX 3090, well inside the competition's practical budget.

TABLE V
MEASURED PQ BY ANNOTATOR COUNT (REFINE PIPELINE).

passes	imgs	measured	SQ	RQ	FN/TP
1	66	0.377	0.672	0.561	1.11
2	16	0.370	0.665	0.556	1.05
3	24	0.437	0.670	0.652	0.68

IX. REPRODUCIBILITY

All code, both model checkpoints (YOLO 20 MB, U-Net 94 MB) and a notebook that loads them without retraining are released with this report. The metric implementation reproduces the organiser's self-evaluation notebook exactly: ground truth is grouped by annotator-image, predictions by physical image, every IoU$>$0.5 pair is a true positive, and PQ is pooled globally. Local numbers in this report use the same mutual-exclusion constraint the platform enforces, so they are comparable to the leaderboard, not merely to each other. The union-oracle and per-group analyses are produced by the same code path, which lets every claim above be re-derived from the released artefacts.

Three practical notes for re-runs: (i) the 601/106 split is keyed by physical image name, so regenerating labels cannot leak a validation image into training; (ii) inference must feed the detector the same 1280 px scale used in training – a 1536 px pass costs 0.008 PQ; (iii) masks are written as compressed COCO RLE and validated for mutual exclusivity before submission, since the platform rejects overlapping masks outright.

X. Conclusion

Local validation PQ is **0.399**; public leaderboard **0.340**. Under the multi-annotator protocol the achievable ceiling on this split is 0.782, and the binding constraints are the small training set, thin low-contrast targets, and FP/FN symmetry that punishes both over- and under-prediction. U-Net-guided proposal refinement is the one component that transfers to the leaderboard (+0.05).